

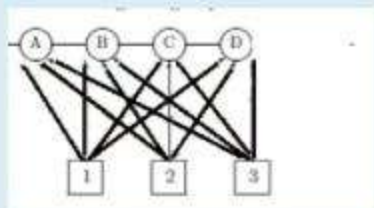
Suppose you have been given a fair coin and you want to find out the odds of getting heads. Which of the following option is true for such a case?

Select one:

- ☒ a. odds will be 1
- ☐ b. odds will be 0.5
- ☐ c. odds will be 0
- ☐ d. None of these



Below is a diagram of a self-organizing map:



Hint: The winner is determined using Euclidean distance.

Suppose that the SOM, shown in Figure, is used to classify types of Cars based on three parameters: Size, speed and passenger load. The weights of the output nodes are shown in the table below:

|        | A | B | C | D |
|--------|---|---|---|---|
| Size   | 3 | 5 | 1 | 2 |
| Speed  | 2 | 1 | 5 | 3 |
| P_Load | 5 | 1 | 1 | 2 |

Each of the three parameters is assessed on a scale from 1 to 5. For example, small cars have size 1, while huge cars would have value 5.

Answer each of the following questions:

a) How many types of cars can this SOM classify?

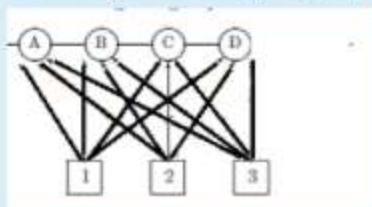
b) Which node will be the winner, if a vector representing airplane x (2; 3; 2) is fed into the input?

c) Suppose you were asked to change the design of the SOM in order to take into account one additional parameters: Price. What would you need to change in this SOM?

increase the number of dimensions in the input space to 4

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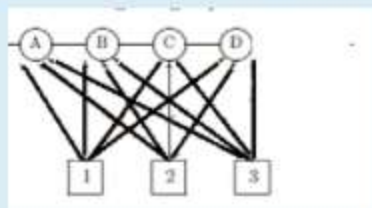
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Answer each of the following questions:

- a) How many types of cars can this SOM classify?
- b) Which node will be the winner, if a vector representing a car with parameters (2; 3; 2) is fed into the input?
- c) Suppose you were asked to change the design of the SOM in order to take into account one additional parameters: Price. What would you need to change in this SOM?  
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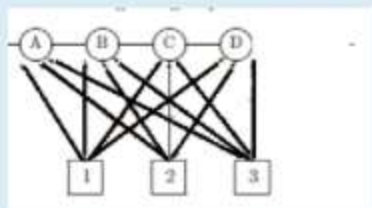
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Answer each of the following questions:

a) How many types of cars can this SOM classify?

b) Which node will be the winner, if a vector representing airplane  $x(2; 3; 2)$  is fed into the input?

c) Suppose you were asked to change the design of the SOM in order to take into account one additional parameters: Price. What would you need to change in this SOM?

- Choose...
- no change
  - increase the number of dimensions in the output space to 6
  - increase the number of dimensions in the input space to 4
  - increase the number of weights to node 1 to 9

A regression between foot length (response variable in cm,  $y$ ) and height (independent variable in inches,  $x$ ) for 33 students resulted in the following regression equation:

$$y = 10.9 + 0.23x$$

One student in the sample was 73 inches tall with a foot length of 29 cm. What is the residual (error) for this student?

Select one:

- ☐ a. 29 cm
- ☒ b. 1.31 cm
- ☐ c. -1.31 cm
- ☐ d. 0.00 cm

A 4-input neuron has weights 1, 2, 3 and 4. The transfer function is linear i.e. ( $f(x)=2 \cdot x$ ). The inputs are 4, 10, 5 and 20 respectively. The output will be:

Select one:

☐ a. 123

☒ b. 238

☐ c. 76

☐ d. 119



Why are linearly separable problems interesting to neural network researchers?

Select one:

- ☒ a. Because they are the only problems that a perceptron can solve successfully.
- ☐ b. Because they are the only problems that a neural network can solve successfully.
- ☐ c. Because they are the only mathematical functions that you can draw.
- ☐ d. Because they are the only mathematical functions that are continuous.

The logit function(given as  $l(x)$ ) is the log of odds function. What could be the range of logit function in the domain  $x=[0,1]$ ?

Select one:

- ☒ a.  $(-\infty, \infty)$
- ☐ b.  $(0, \infty)$
- ☐ c.  $(0, 1)$
- ☐ d.  $(-\infty, 0)$



In supervised learning, training set of data includes

Select one:

- ☐ a. Input
- ☒ b. Both input and output
- ☐ c. None
- ☐ d. Output

In machine learning the simple linear regression equation can be written as  $y = b + wx$   
the parameter  $b$  represents the

Select one:

- ☐ a. estimated weight
- ☐ b. independent variable
- ☐ c. average or predicted response
- ☒ d. estimated bias



Which statement is not true about gradient descent?

Select one:

- ☒ a. During computation of the gradient descent the steps are taken proportional to the positive of the gradient (or approximate gradient) of the function at the current point.
- ☐ b. It is related to derivative of a function.
- ☐ c. It is used in many AI algorithms to optimize the cost function.
- ☐ d. Gradient descent is a first-order iterative optimization algorithm for finding a local minimum of a differentiable function.

## Advantages of Rectified Linear Unit (ReLU) activation function:

Select one:

- ☒ a. has simple calculation.
- ☐ b. All mentioned
- ☐ c. supports faster convergence in Stochastic gradient descent.
- ☐ d. Effectively alleviates the vanishing gradient.

Backpropagation is the transfer of error back through the network using a set of recurrent connections.

Select one

☒ True

☐ False



What is the name of this principle in machine learning: "The goal of machine learning is that the learnt model should perform well on new samples, not just those on which the model has been trained"

Select one:

- ☐ a. Underfitting
- ☒ b. Generalization
- ☐ c. Overfitting
- ☐ d. Classification



Neural network training is accomplished by repeatedly passing the training data through the network while

Select one:

- ☐ a. training instance attribute values are modified.
- ☒ b. individual network weights are modified.
- ☐ c. individual network nodes have the coefficients on their corresponding functional parameters modified.
- ☐ d. the ordering of the training instances is modified.

The following data: (1, 1), (20, 400), (30, 800), (40, 1300) is regressed with least squares regression to  $y = b + wx$ . The value of  $w$  most nearly is

Select one:

- ☒ a. 32.625
- ☐ b. 40.000
- ☐ c. 27.480
- ☐ d. 28.956



SOM is a supervised learning algorithm neural network.

Select one

☐ True

☒ False



An example of a hyperparameter of a neural network:

Select one:

- ☒ a. Learning rate
- ☐ b. Weights
- ☐ c. Bias
- ☐ d. Number of inputs



We have trained a machine learning model to identify that whether the object in an image is a cat. Now we use 200 images to test its performance. Among 200 images, 180 contain cats and 20 do not contain cats. The identification result of the model is that 140 images contain cats and 60 do not contain cats. Given that TP (true positive) of the model = 135.

What is the Accuracy of the model?

Select one:

- ☐ a. 96.4 %
- ☐ b. 82.5 %
- ☐ c. 75.7 %
- ☒ d. 75 %

